Weighted Omics View Embedding via Nystrom (WOVEN): Supervised Multi-Omics Integration for Block-Missing Clinical Cohort Data

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Abstract

Motivation: Supervised multi-omics integration enforces a complete-case constraint, silently discarding patients missing any modality. When missingness correlates with disease severity, complete-case analysis can exclude the most affected patients and bias downstream inference.

Results: We present WOVEN (Weighted Omics View Embedding via Nystrom), which learns a shared supervised latent space from fully observed anchor subjects and projects incomplete patients by a linear out-of-sample (Nystrom) extension without feature-level imputation. Across 400 simulation replicates, WOVEN retains 100% of subjects at 50% block-missingness versus 19% for complete-case supervised methods, achieves three-fold higher silhouette on complete data, and lower classification error on compositional microbiome-metabolomics data. Feature-level imputation before integration does not recover lost cross-modal structure. Applied to 2,422 ADNI subjects combining MRI, lipidomics, and metabolomics, WOVEN scores 70% of patients versus 31%; recovered patients show 25% dementia prevalence versus 12%.

Availability: R package: https://github.com/NathanBresette/woven (Bioconductor submission in progress). Benchmark: https://github.com/NathanBresette/woven_paper.

1. Introduction

1. The Context and The Bottleneck

2. The Gap in Existing Tools

3. The Proposed Method

4. The Validation Summary

2. Methods

2.1 Problem Formulation

Let $X^{(v)} \in \mathbb{R}^{n \times p_v}$ denote the data matrix for modality $v \in \{1, \ldots, V\}$ over n subjects. Under block-missing data, subject i may be entirely absent from modality v. Let $\mathcal{A} \subseteq \{1, \ldots, n\}$ denote the anchor set, the subjects with observations in all V modalities, with $n_a = |\mathcal{A}|$, and let $X_a^{(v)} \in \mathbb{R}^{n_a \times p_v}$ denote the restriction of $X^{(v)}$ to anchor rows. Non-anchor subjects have data in at least one but not all modalities. Subjects with no data in any modality are excluded. Let $Y \in \{1, \ldots, C\}^n$ denote the vector of class labels and $K \leq n_a$ the number of latent dimensions.

WOVEN seeks projection matrices $W^{(v)} \in \mathbb{R}^{p_v \times K}$ for each modality such that the anchor latent scores $Z_a^{(v)} = X_a^{(v)} W^{(v)}$ are maximally correlated across modalities, geometrically regularized within each modality, and discriminative with respect to Y.

2.2 Objective Function

The WOVEN objective is a label-augmented multiset canonical correlation analysis (MCCA), the multiview extension of canonical correlation analysis (CCA), maximizing a sum-of-correlations (SUMCOR) criterion:

$$\max_{W^{(v)}} \sum_{v<u} \operatorname{tr}\left(W^{(v)T} \tilde{C}_{vu} W^{(u)}\right) \quad \text{s.t.} \quad W^{(v)T} B_v W^{(v)} = I_K$$

where $\tilde{C}_{vu} = X_a^{(v)T} \left(I + \gamma_Y K_Y\right) X_a^{(u)}$ is the label-augmented cross-covariance restricted to anchor subjects, $\tilde{Y} \in \mathbb{R}^{n_a \times C}$ is the column-centered indicator matrix of anchor class labels, $K_Y = \tilde{Y}\tilde{Y}^T / n_a$ is the resulting label kernel (entries large when subjects share the same class, zero-mean otherwise), and $\gamma_Y \geq 0$ is the supervision strength hyperparameter.

The constraint matrix $B_v = X_a^{(v)T} M_v X_a^{(v)}$ encodes graph Laplacian regularization, which preserves local neighborhood geometry in the learned embedding (Belkin and Niyogi, 2003), via $M_v = I_{n_a} + \lambda_v L_a^{(v)} / n_a$, where $L_a^{(v)}$ is the $n_a \times n_a$ anchor submatrix of a k-nearest-neighbor radial basis function (RBF) graph Laplacian built from all observed rows of $X^{(v)}$ (block-missing subjects excluded from the k-NN graph, no imputation). Restricting B_v to the anchor block ensures that non-anchor subjects do not contribute to parameter estimation; the graph is still built on the full observed cohort, preserving neighborhood structure across all available data.

2.3 Closed-Form Solution via Dual MCCA

The WOVEN objective is reformulated in dual (sample) space using M_v and its symmetric square root $M_v^{1/2}$ as defined in Section 2.2, enabling a single eigendecomposition of a $(Vn_a) \times (Vn_a)$ block matrix. The dual block matrix P has zero diagonal blocks and off-diagonal blocks:

$$P_{vu} = M_v^{-1/2} \left(I + \gamma_Y K_Y\right) M_u^{-1/2}$$

The top-K eigenvectors of P (positive eigenvalues only), partitioned into V blocks $\Phi_v \in \mathbb{R}^{n_a \times K}$, give M-orthonormalized anchor latent scores $Z_v = M_v^{-1/2}\Phi_v$; projection matrices are then recovered as $W^{(v)} = X_a^{(v)T}(X_a^{(v)} X_a^{(v)T} + \epsilon I)^{-1} Z_v$, with a small ridge $\epsilon > 0$ for numerical stability. The eigendecomposition is $O((Vn_a)^3)$, and setting $\gamma_Y = 0$ recovers unsupervised SUMCOR MCCA. The full derivation, the M-orthonormalization, the reduction to dual supervised CCA at $V = 2$, and the treatment of the ridge and of negative eigenvalues are given in Supplementary Methods S2.

2.4 Projection of Block-Missing Subjects

All non-anchor subjects, regardless of how many modalities are observed, are projected into the latent space via linear out-of-sample extension using the estimated projection matrices:

$$\hat{z}_i = \frac{1}{|\mathcal{V}_i|} \sum_{v \in \mathcal{V}_i} x_i^{(v)} W^{(v)}$$

where $\mathcal{V}_i$ is the set of observed modalities for subject i and $|\mathcal{V}_i| \geq 1$. Absent modalities contribute nothing to $\hat{z}_i$; no feature-level imputation is performed. For subjects with a single observed modality, the formula reduces to $\hat{z}_i = x_i^{(v)} W^{(v)}$. The application of estimated projection matrices to out-of-sample subjects follows the Nystrom out-of-sample extension framework (Bengio *et al.*, 2003), wherein $W^{(v)}$ learned from anchor subjects serves as the linear operator mapping new observations into the anchor latent space. For linear embeddings such as those in the CCA family, this out-of-sample extension reduces exactly to applying the learned projection operator to new feature vectors, requiring no kernel evaluation against anchor subjects.

Algorithm 1 (WOVEN). *Input:* per-modality matrices $X^{(v)}$, labels Y, anchor set $\mathcal{A}$, latent dimension K, hyperparameters λ_v, γ_Y, k.

1. Build a k-NN RBF graph Laplacian $L^{(v)}$ on the observed rows of each modality (block-missing subjects excluded); form $M_v = I + \lambda_v L_a^{(v)}/n_a$ on the anchor block.
2. Form the label kernel K_Y from anchor labels and assemble the dual block matrix P with off-diagonal blocks $M_v^{-1/2}(I + \gamma_Y K_Y)M_u^{-1/2}$ and zero diagonal blocks.
3. Take the top-K positive eigenvectors of P; the v-th block gives anchor latent scores Z_v (M-orthonormalized).

4. Recover projection matrices $W^{(v)} = X_a^{(v)T}(X_a^{(v)} X_a^{(v)T} + \epsilon I)^{-1} Z_v$.
5. Project each non-anchor subject as the mean of $x_i^{(v)} W^{(v)}$ over its observed modalities.

Output: projection matrices $W^{(v)}$ and latent scores for all subjects.

2.5 Hyperparameter Selection

The supervision strength γ_Y is selected from $\{0.5, 1.0, 5.0, 10.0\}$ and the regularization strength λ_v from $\{0.001, 0.005, 0.01, 0.1, 0.5\}$ by three-fold stratified cross-validation on anchor subjects, maximizing silhouette score (Rousseeuw, 1987) of held-out anchor latent scores. The k-NN graph uses $k = 10$ by default; sensitivity analysis across $k \in \{3, 5, 10, 20, 50\}$ shows silhouette varies by less than 0.044 across this range. Sensitivity to λ across four orders of magnitude (10^{-4} to 1.0) changes silhouette by less than 1%, and any $\gamma_Y > 0$ recovers full performance (setting $\gamma_Y = 0$ degrades silhouette from 0.799 to -0.094, confirming that the supervision term is essential). Given this stability, the R package defaults to these values ($k = 10$, $\lambda_v = 0.1$, $\gamma_Y = 1$), so cross-validation is optional rather than required for routine use.

2.6 Simulation Design

All benchmark data are semi-synthetic: distributional parameters were estimated from real reference datasets before WOVEN was developed, so the simulation is independent of method construction (Sankaran *et al.*, 2025). Each arm has $n = 300$ subjects in four balanced groups of 75, with a shared $K = 5$ latent-factor ground truth embedded across modalities by SUMO (Osang'ir *et al.*, 2025) for evaluating latent-space recovery. Four arms span distinct signal regimes (full simulator specifications in Supplementary Methods S1):

- **ARM A** (RNA-seq + methylation, $V = 2$): diffuse signal; no single modality cleanly separates the four groups. 4,047 genes (SPsimSeq; Assefa *et al.*, 2020) and 367 CpG sites (InterSIM; Chalise *et al.*, 2016).
- **ARM B** (RNA-seq + methylation + proteomics, $V = 3$): diffuse signal extended with 62 proteins; the third modality lowers the anchor fraction to approximately 13% at a 50% block-missing rate.
- **ARM C** (RNA-seq + methylation + proteomics, $V = 3$): concentrated signal with strong between-group separation, where any reasonable method should succeed.
- **ARM D** (microbiome + metabolomics, $V = 2$): compositional, zero-inflated signal with spiked-in cross-modal associations (MIDASim, He *et al.*, 2024; NorTA, Mangnier *et al.*, 2025; SparseDOSSA2, Ma *et al.*, 2021); microbiome features are centered log-ratio (CLR) transformed (Aitchison, 1982).

Block missingness is induced post-simulation (Rubin, 1976): entire modality blocks are removed per subject at 30% and 50% missing completely at random

(MCAR), and under structured missing at random (MAR; missingness probability correlated with group label), with every subject retaining at least one observed modality. One hundred independent replicates per arm yield 400 total per method per condition. In every replicate the anchor set (subjects observed in all V modalities) is the training set: under MCAR 50%, anchor fractions are approximately 25% ($n_a \approx 75$, $V = 2$) and 13% ($n_a \approx 39$, $V = 3$). All projection matrices $W^{(v)}$ and hyperparameters are estimated from anchors alone; non-anchor subjects are scored after fitting. Simulation scripts are at https://github.com/NathanBresette/woven_paper.

2.7 Benchmark Evaluation Protocol

We evaluated WOVEN against DIABLO (Singh *et al.*, 2019; Rohart *et al.*, 2017), MOFA+ (multi-omics factor analysis; Argelaguet *et al.*, 2020), ImputeDIABLO (missForest random-forest imputation, Stekhoven and Bühlmann, 2012, followed by DIABLO), IntegrAO (Ma *et al.*, 2025), and dense DIABLO (DIABLO with sparsity constraint removed, evaluated on ARM D only to isolate whether DIABLO's disadvantage is algorithmic or a sparsity artifact). All methods were run with $K = 5$ latent dimensions. The benchmark structure, comparing integration methods across simulated multi-omics arms under controlled signal conditions, follows the design precedent of prior multi-omics integration comparisons (Duan *et al.*, 2021). The simulation parameters were fixed from real reference data before WOVEN was developed (Section 2.6), and the arms span both favorable and unfavorable regimes for WOVEN, from concentrated signal to diffuse-signal arms where no method exceeds chance, and from MCAR to MAR missingness where the anchor set is a biased subsample. ADNI (Section 2.8) serves as the independent test set.

Balanced error rate (BER) is computed via per-fold dimensionality reduction (DR) refitting with linear discriminant analysis (LDA), ensuring test subjects never contribute to parameter estimation; this avoids the circularity of fixed-Z BER for supervised methods. For IntegrAO, which produces graph embeddings rather than projection matrices, held-out subjects are projected via k-nearest-neighbor weighted average of training embeddings in feature space using available views, then classified by LDA, matching the per-fold DR refitting protocol used for all other methods. All methods use identical three-fold splits per replicate. Chance BER is 0.75 for four balanced classes. Latent space geometry is assessed using silhouette score (Rousseeuw, 1987) and normalized mutual information (NMI). For methods that score only a subset of subjects (DIABLO), full-cohort and anchor-only silhouette scores are both reported to enable fair comparison. All reported values are means across 100 replicates per arm; high cross-condition variability in BER reflects between-arm signal differences (ARM C: BER = 0.000 vs ARM A: BER $\approx$ 0.67) rather than estimation uncertainty. Per-arm standard deviations are provided in Supplementary Table S1. Differences between WOVEN and DIABLO were assessed by the Wilcoxon signed-rank test, pairing replicates by arm and replicate index within each condition; WOVEN's lower BER and higher anchor silhouette are significant in every condition (each $p < 10^{-14}$; $n = 400$ paired replicates).

2.8 ADNI Validation

We applied WOVEN to 2,422 baseline subjects from the Alzheimer's Disease Neuroimaging Initiative (ADNI; Jack *et al.*, 2008) combining three modalities: MRI (magnetic resonance imaging) FreeSurfer morphometrics (341 features, $n_{\text{obs}} = 902$), plasma lipidomics (781 features, $n_{\text{obs}} = 1,418$), and NMR (nuclear magnetic resonance) metabolomics (267 features, $n_{\text{obs}} = 1,642$). The anchor set (complete across all three modalities) contains 743 subjects (31%). ADNI is used here as a methodological testbed for block-missing multi-omics integration; we make no claims about the biological specificity of plasma lipids for central nervous system pathology. The three modalities were selected to maximize subject coverage and to span distinct missingness mechanisms: MRI requires a separate imaging visit and is therefore missing independently of the blood-based assays, whereas plasma lipidomics and NMR metabolomics derive from a dedicated blood draw. Lipidomics and NMR share that draw and so have correlated missingness, but are measured on independent analytical platforms. Cerebrospinal fluid (CSF)-based biomarkers, while biologically proximal to neurodegeneration, are available for fewer than 300 ADNI subjects and would substantially reduce the recoverable cohort, undermining the effective sample size (ESS) demonstration. Labels are baseline diagnosis: cognitively normal (CN), mild cognitive impairment (MCI), and dementia (three-class; chance BER = 0.667). Silhouette and NMI on ADNI were computed on each method's fitted (in-sample) latent scores; balanced error rate was computed on the shared 743-subject anchor set by five-fold per-fold DR refitting, identical to the non-circular simulation protocol (Section 2.7), with both methods refit on training anchors within each fold. Data were obtained from the ADNI database (adni.loni.usc.edu). ADNI is funded by the National Institute on Aging and the National Institute of Biomedical Imaging and Bioengineering. A complete list of ADNI investigators is at adni.loni.usc.edu/about/faqs.

3. Results

WOVEN's defining property is that it scores every enrolled subject. Effective sample size stays at 1.00 under all missingness levels, while the standard supervised method DIABLO retains only 19% of subjects at 50% missingness (Section 3.3). This retention does not cost accuracy. WOVEN's balanced error rate matches or beats DIABLO in every condition and is far lower on compositional microbiome-metabolomics data (Sections 3.1, 3.3), feature-level imputation does not close that gap (Section 3.4), and as modalities go missing WOVEN degrades more gracefully than IntegrAO, the closest block-missing competitor (Section 3.2). On complete data it also roughly triples DIABLO's latent-space quality (silhouette 0.83 versus 0.27; Section 3.1). In our benchmarks, then, the complete-case constraint cost subjects, disproportionately the most affected ones

(Section 3.5), and bought no accuracy in return. Tables 1 to 3 and Figures 1 to 3 summarize these results.

3.1 WOVEN Recovers Latent Structure More Precisely on Complete Data

On complete data, WOVEN recovers the cleanest latent geometry of any method tested (Table 1, Figure 2A), roughly tripling DIABLO's silhouette (0.828 vs 0.271) and reaching perfect cluster recovery (NMI = 1.00 in every replicate of every arm). This follows directly from the label-augmented objective, which aligns the canonical directions with class structure. One caveat governs how to read these two metrics: both are computed on the embedding each method was fit on, so they measure how cleanly the latent space encodes the labels, not how well it generalizes; generalization is the job of the held-out BER. The two can part ways. On the diffuse-signal arms A and B the groups are present (NMI = 1.00) but too weakly separated for held-out classification (BER near chance), and no method recovers signal there.

Classification error tells a sharper, arm-specific story (Table 2). The overall complete-data gap (BER 0.375 for WOVEN, 0.528 for DIABLO) is carried almost entirely by ARM D, the compositional microbiome-metabolomics arm, where WOVEN's 0.145 sits far below DIABLO's 0.763. Removing DIABLO's sparsity constraint barely moves it (0.762 over 100 replicates), so this is an algorithmic gap, not a sparsity artifact. On the concentrated-signal ARM C both methods are perfect.

WOVEN runs in 3.2 to 12.6 seconds per replicate under complete-data conditions (ARM D fastest, ARM B slowest), compared to 4.3 to 72.8 seconds for DIABLO and 10.2 to 16.8 seconds for MOFA+. WOVEN is 5.8 to 6.8-fold faster than DIABLO on the high-dimensional ARM A and B arms (ARM B: 12.6 vs 72.8 s; ARM A: 10.6 vs 72.3 s).

Table 1. Subject retention and latent-space geometry across 400 replicates (4 arms $\times$ 100 reps per condition). Sil: silhouette over all scored subjects; Sil (anchor): silhouette over anchor subjects only. ESS: fraction of enrolled subjects scored. ImputeDIABLO applies only under missingness. Bold marks the best value per metric per condition on a matched population; full-cohort NMI under missingness is left unbolded (Section 3.3). Balanced error rate is reported by arm in Table 2.

Condition	Method	Silhouette	Sil (anchor)	NMI	ESS
Complete	WOVEN	**0.828**	0.828	**1.000**	1.00
Complete	DIABLO	0.271	0.271	0.539	1.00
Complete	MOFA+	0.204	0.204	0.500	1.00
MCAR 30%	WOVEN	0.293	**0.776**	0.535	**1.00**
MCAR 30%	DIABLO	0.298	0.298	0.592	0.42
MCAR 30%	MOFA+	0.189	0.197	0.475	1.00

Condition	Method	Silhouette	Sil (anchor)	NMI	ESS
MCAR 30%	ImputeDIABLO	0.209	0.271	0.499	1.00
MCAR 50%	WOVEN	0.218	**0.710**	0.476	**1.00**
MCAR 50%	DIABLO	0.350	0.350	0.739	0.19
MCAR 50%	MOFA+	0.179	0.192	0.462	1.00
MCAR 50%	ImputeDIABLO	0.195	0.267	0.498	1.00
MAR	WOVEN	0.297	**0.757**	0.546	**1.00**
MAR	DIABLO	0.313	0.313	0.597	0.41
MAR	MOFA+	0.186	0.208	0.473	1.00
MAR	ImputeDIABLO	0.201	0.280	0.495	1.00

Table 2. Balanced error rate by simulation arm, on complete data and at 50% MCAR (per-fold DR refitting with LDA; four-class chance = 0.75). ImputeDIABLO applies only under missingness; IntegrAO is compared in Section 3.2. Bold marks WOVEN values that improve on DIABLO.

Arm	Complete WOVEN	Complete DIABLO	MCAR 50% WOVEN	MCAR 50% DIABLO	MCAR 50% ImputeDIABLO
A (RNA+meth)	0.673	0.674	0.713	0.715	0.739
B (RNA+meth+prot)	0.682	0.675	0.758	0.754	0.762
C (InterSIM)	0.000	0.000	**0.007**	0.040	0.040
D (micro+metab)	**0.145**	0.763	**0.113**	0.777	0.777
All arms (mean)	**0.375**	0.528	**0.398**	0.571	0.579

3.2 ARM D Reversal: Nystrom Projection Is More Robust to Block-Missingness Than Graph Diffusion

Comparing WOVEN with IntegrAO shows where its block-missing advantage actually comes from. IntegrAO was evaluated in a dedicated 100-replicate benchmark on the same simulation arms; all WOVEN numbers in this subsection are taken from that same benchmark run for a direct comparison. WOVEN's ARM D BER therefore differs slightly here (0.131 complete, 0.107 at 50% MCAR) from the main-benchmark values reported elsewhere (0.145 and 0.113; Sections 3.1, 3.3), reflecting independent cross-validation fold seeds; the difference is within replicate-level variation. On ARM D complete data, IntegrAO achieves BER = 0.086 versus WOVEN 0.131, confirming that unsupervised graph diffusion better captures NorTA microbiome-metabolomics structure when all data are available. Under 50% MCAR, this reverses: WOVEN BER is 0.107 versus IntegrAO 0.130. IntegrAO's graph diffusion constructs a subject-by-subject affinity matrix; when 50% of subjects are missing a modality, the affinity matrix is partially unobserved and must be estimated, introducing noise that degrades embedding quality. WOVEN's Nystrom projection operates on the anchor-estimated eigenvector structure, requiring only available-modality features per new subject, with no dependence on the full affinity matrix. The ARM D reversal demonstrates WOVEN's mechanism directly: anchor-restricted estimation followed by closed-form projection degrades more gracefully than graph-based methods when modality availability is heterogeneous.

IntegrAO and WOVEN achieve equivalent performance on ARM C (BER = 0.000 for both) and WOVEN outperforms IntegrAO on diffuse arms (WOVEN 0.673/0.680 vs IntegrAO 0.747/0.763 complete), where label supervision provides a consistent advantage. IntegrAO also achieves ESS retention of 1.00 in all conditions, matching WOVEN on subject retention. The ESS advantage in this benchmark is WOVEN and IntegrAO together versus DIABLO.

3.3 WOVEN Retains All Subjects Under Block-Missingness

At 50% MCAR block-missing rates, DIABLO retains 19% of subjects by enforcing the intersection constraint. WOVEN retains 100% of subjects in every arm and condition. MOFA+ also achieves ESS = 1.00 via variational Bayes (VB) inference that skips missing entries. The distinction between WOVEN and MOFA+ under missingness lies in supervised geometry: WOVEN maintains anchor-only silhouette 0.710 at 50% MCAR versus MOFA+ 0.192, and provides BER estimates via per-fold DR refitting with labeled data, whereas MOFA+'s unsupervised objective does not.

DIABLO's apparent full-cohort silhouette advantage under missingness (0.350 vs WOVEN 0.218) is an artifact of selection: DIABLO only scores the 19% of subjects with complete data, who cluster more cleanly by construction. The anchor-only comparison, evaluating both methods on the same population, reverses this decisively (WOVEN 0.710 vs DIABLO 0.350; Figure 2A). The same selection mechanism explains DIABLO's higher full-cohort NMI under missingness (Table 1): DIABLO's geometry metrics reflect only its retained complete-case subjects, whereas WOVEN's full-cohort values additionally include projected subjects, many with a single observed modality and correspondingly noisier latent estimates.

BER at 50% MCAR is 0.398 for WOVEN and 0.571 for DIABLO overall, driven by ARM D (0.113 vs 0.777; Table 2). MAR block-missingness produces comparable results (WOVEN BER 0.389 vs DIABLO 0.572), so the advantage holds under both missingness mechanisms.

Notably, WOVEN runs faster under block-missingness than on complete data: 1.6 to 6.2 seconds per replicate at 50% MCAR, versus 3.2 to 12.6 seconds on complete data. This speedup arises because the anchor set is smaller under missingness, reducing the eigendecomposition from $O((Vn)^3)$ on complete data to $O((Vn_a)^3)$ where $n_a \approx 0.25n$ for $V = 2$ and $n_a \approx 0.13n$ for $V = 3$ at 50% MCAR. DIABLO also speeds up under missingness (31.8 to 38.1 seconds for ARM A/B at mcar50 vs 72 seconds complete), for the same reason: it fits only on anchor subjects. WOVEN remains 5 to 7-fold faster than DIABLO at 50% MCAR on the high-dimensional arms, while scoring the full cohort rather than the complete-case subset.

3.4 Imputation Does Not Recover Discriminative Signal

Imputing the missing blocks before integration does not help. ImputeDIABLO (missForest imputation followed by DIABLO) tracks DIABLO almost exactly and stays well behind WOVEN in every condition (Tables 1, 2): at 50% MCAR its overall BER (0.579) matches DIABLO's (0.571), and on ARM D the two are identical (0.777), far above WOVEN's 0.113. Its anchor silhouette never reaches even DIABLO's complete-case value.

Imputation and latent-space projection differ in a way that matters here. missForest predicts individual molecular feature values from within-modality cross-feature correlations, then passes the imputed matrix to DIABLO. Imputation does not reconstruct the cross-modal alignment structure that block-missingness disrupts; it replaces missing entries with within-modality predictions that carry no information about how that modality relates to others. WOVEN projects incomplete subjects through the estimated cross-modal projection matrices $W^{(v)}$, preserving the geometric relationships learned from anchor subjects.

3.5 ADNI: WOVEN Recovers Clinically Relevant Subjects That DIABLO Discards

The simulations establish that WOVEN retains the full cohort without sacrificing accuracy under designed MCAR and MAR missingness. In a real cohort the stakes are higher, because missingness is driven by clinical mechanisms rather than a removal probability, and the subjects who go missing are not a random subset.

Applied to 2,422 ADNI subjects, WOVEN scores 1,687 (70%) to DIABLO's 743 (31%), recovering 944 patients DIABLO discards (Table 3). On the shared anchor set, under the same per-fold protocol as the simulations, WOVEN's BER is lower than DIABLO's (0.546 vs 0.642; three-class chance = 0.667), and its in-sample silhouette and NMI are both higher (Figure 3B). Classification here is near chance for both methods, as expected for blood-and-imaging biomarkers of a heterogeneous clinical syndrome; the point of the ADNI analysis is subject coverage on real cohort data, which is where the two methods diverge sharply.

The recovered subjects differ systematically from those retained (Table 3, Figure 3A): dementia prevalence among them is 25%, twice the 12% among DIABLO-retained subjects, and the largest recoverable group (660 subjects with lipidomics and NMR but no MRI) is missing only the imaging modality. A comparative effectiveness research (CER) analysis built on a DIABLO-derived latent space would therefore underrepresent the most severely affected patients, biasing downstream subgroup estimates.

Table 3. ADNI subject retention and baseline diagnosis distribution by group. The 735 subjects with no observed modality cannot be scored by any method.

Group	N	CN	MCI	Dementia
DIABLO retained (complete across all 3 modalities)	743	34%	54%	12%
WOVEN-only (missing at least 1 modality)	944	28%	47%	**25%**
Unscored (no data in any modality)	735	53%	36%	11%
All enrolled	2,422	38%	47%	15%

3.6 Robustness to Anchor Fraction and Anchor Selection

WOVEN estimates all parameters from anchors, so we characterized how performance depends on the anchor fraction (Supplementary Figure 2; anchor fraction swept from 5% to 75%, 50 replicates per setting, ARMs A and C). The shared latent space is recoverable from very few complete cases: anchor silhouette remains high and stable across the entire range (ARM A: 0.55 to 0.82; ARM C: 0.58 to 0.89), including at a 5% anchor fraction where only $n_a \approx 16$ subjects are complete. Out-of-sample projection is the more fraction-sensitive component: under diffuse signal (ARM A), single-modality non-anchor subjects are placed near the origin (non-anchor silhouette slightly negative at all fractions), whereas under concentrated signal (ARM C) they are well-placed (non-anchor silhouette 0.30 to 0.62), consistent with the limitation that single-modality subjects receive no cross-modal averaging.

The MAR condition tests anchor composition directly: missingness is correlated with class, so the anchor set is a biased subsample rather than a random one. WOVEN's MAR performance matches its MCAR 50% performance (BER 0.389 vs 0.398; anchor silhouette 0.757 vs 0.710).

Figure Legends

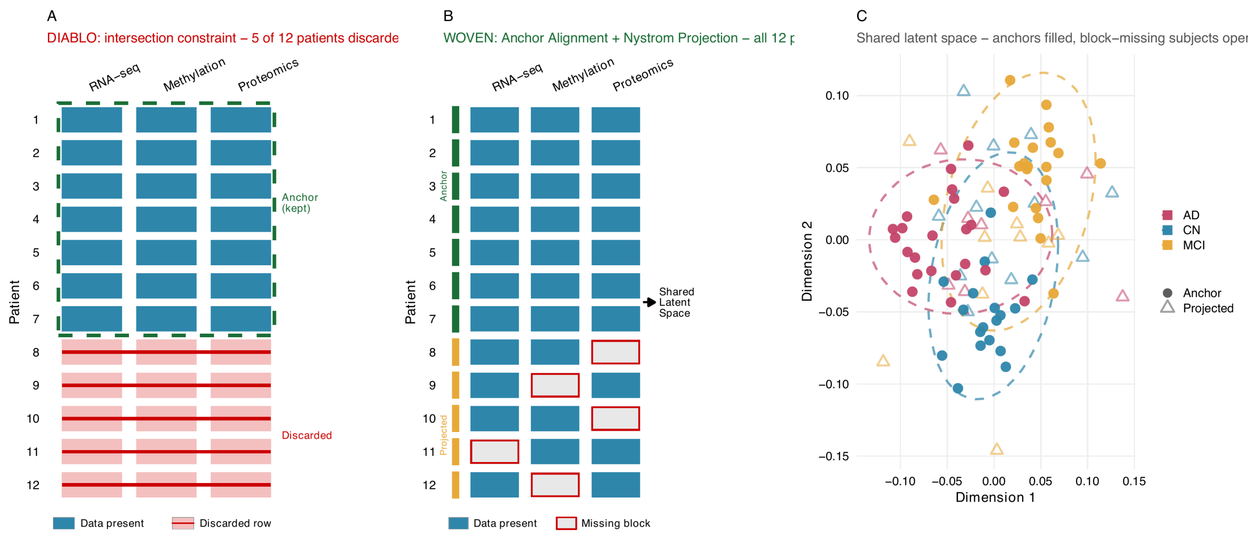

Figure 1. Conceptual overview of WOVEN. **(A)** Complete-case methods discard any subject missing a modality (5 of 12 shown removed). **(B)** WOVEN estimates projection matrices on fully observed anchor subjects (green) and projects block-missing subjects (amber) through their available modalities, so all 12 are scored. **(C)** Shared latent space on the `woven_example` dataset (n=90, 3 modalities, ~33% block-missing); anchors (filled circles) and projected subjects (open triangles) share the same class structure.

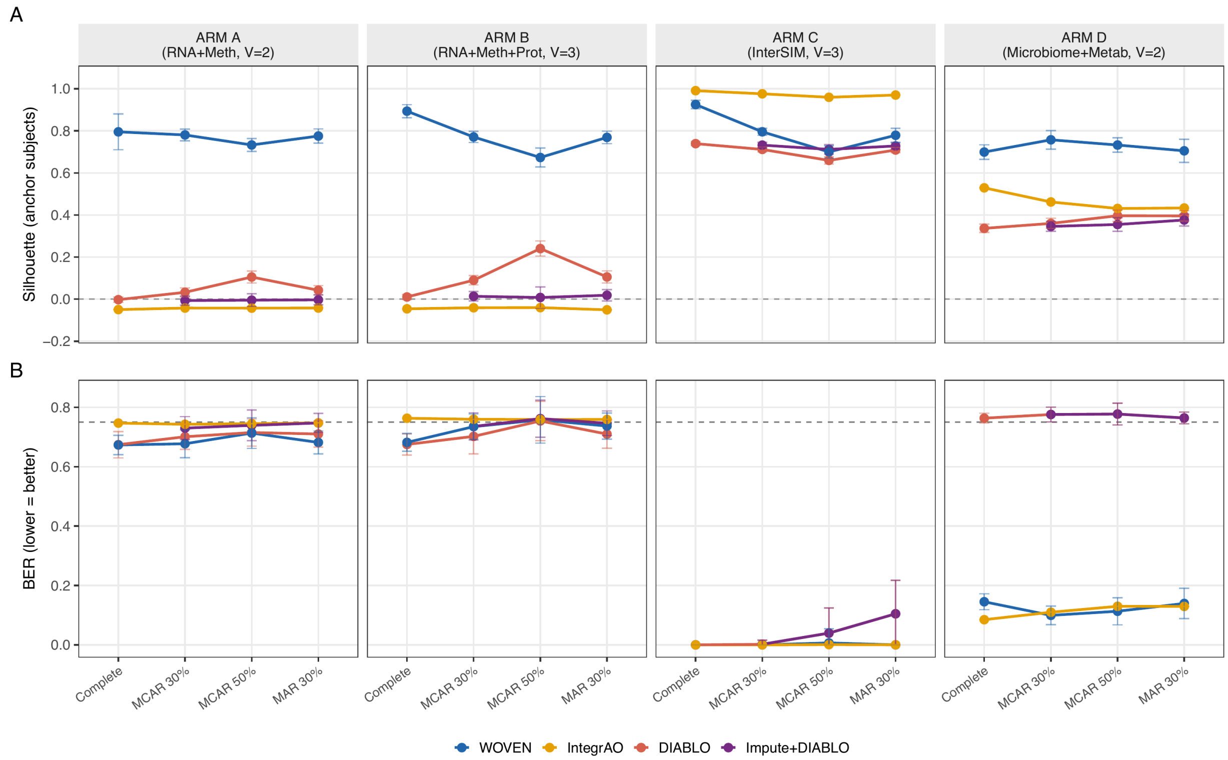

Figure 2. Benchmark across 400 replicates (4 arms $\times$ 100 reps). **(A)** Anchor-only silhouette by condition and arm: WOVEN stays high (0.71–0.90) under all missingness; DIABLO and ImputeDIABLO stay near zero on diffuse arms (A, B); IntegrAO is high only on ARM C. **(B)** BER by condition and arm (per-fold LDA for WOVEN/DIABLO/ImputeDIABLO, k-NN projection for IntegrAO; dashed line = chance, 0.75). Error bars: SD over 100 replicates. Exact BER values are in Table 2.

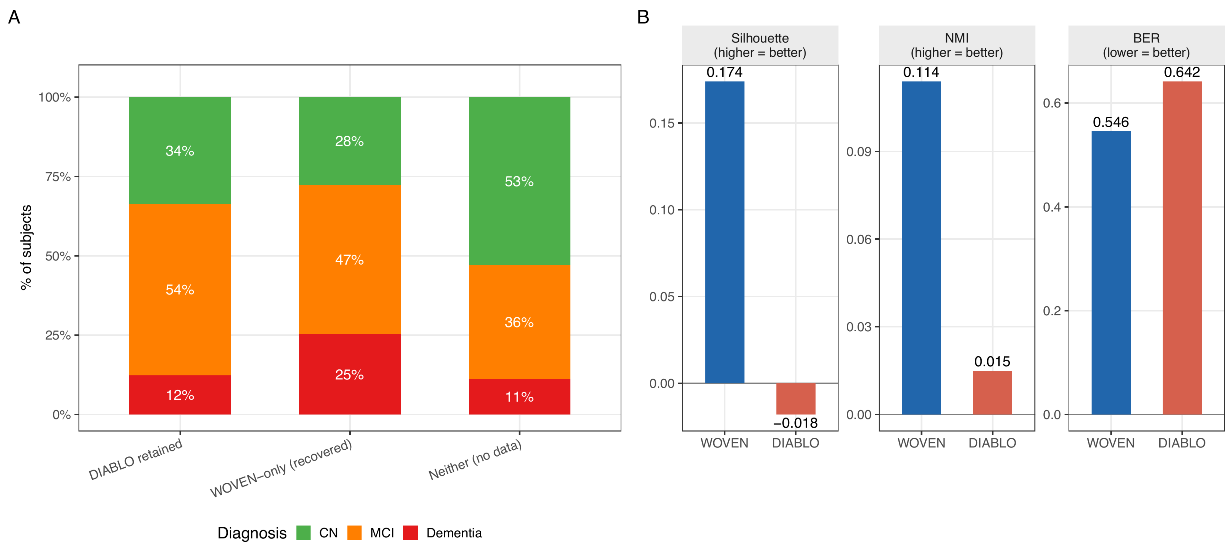

Figure 3. ADNI validation (n=2,422; MRI FreeSurfer + plasma lipidomics + NMR metabolomics). **(A)** Diagnosis distribution by group: DIABLO retains 743 subjects (31%), WOVEN scores 1,687 (70%); recovered subjects have 25% dementia prevalence versus 12% among retained. **(B)** WOVEN versus DIABLO on ADNI: in-sample silhouette (0.174 vs -0.018), NMI (0.114 vs 0.015), and anchor-set BER under per-fold refitting (0.546 vs 0.642; three-class chance = 0.667).

4. Discussion

WOVEN is a supervised multi-omics integration framework that natively handles block-missing cohorts without feature-level imputation. Its central idea is to learn the shared latent space once, on the reliable complete cases, and then score incomplete patients by extending that embedding through their available modalities, rather than first reconstructing their missing measurements. This retains the full cohort. The complete-case constraint of existing supervised methods is therefore an architectural choice, not a technical necessity: it can be relaxed without sacrificing discriminative power or interpretability. The underlying dual SUMCOR MCCA solution is the global optimum of the relaxed trace objective, obtained in a single eigendecomposition rather than by iterative optimization.

On complete data, WOVEN's gain in latent-space geometry over DIABLO reflects the label-augmented objective aligning canonical directions with class-discriminative cross-modal correlations. Its large margin on the compositional ARM D indicates that this label-augmented cross-covariance with Laplacian regularization captures correlation structure that DIABLO's partial least squares (PLS) objective misses. The imputation baseline (ImputeDIABLO) does not recover the discriminative structure that block-missingness disrupts, because within-modality prediction cannot rebuild cross-modal alignment. And the reversal against IntegrAO on ARM D, where unsupervised graph diffusion wins on complete data but loses once modalities go missing, isolates the advantage of anchor-restricted eigendecomposition with closed-form projection over graph diffusion when modality availability is heterogeneous. The three pieces of WOVEN do different work. Label supervision drives the discriminative gains; remove it and performance collapses (Section 2.5). What lets WOVEN score the subjects DIABLO discards is the anchor-restricted estimation and closed-form projection. The graph Laplacian matters least of the three: silhouette changes by under 1% as its strength varies across four orders of magnitude (Section 2.5), so in practice it needs no tuning.

What is new in WOVEN is the combination, not any single component. Prior graph-regularized CCA methods either require fully paired subjects (Graph MCCA; Chen *et al.*, 2019) or, like the semi-supervised LapKCCA (Blaschko *et al.*, 2011), were developed for paired neuroimaging without block-missing support or out-of-sample projection. OLFG (Chen *et al.*, 2023) is the closest structural analog, also pairing graph Laplacian regularization with linear projection matrices, but it imputes features before projecting, whereas WOVEN operates entirely in the latent space. Deep generative approaches (MIMIR, Nambiar *et al.*, 2026; JASMINE, Ballard *et al.*, 2025) reconstruct raw molecular features through decoder networks, performing feature-level imputation and yielding embeddings without interpretable projection matrices. WOVEN's direct predecessor is DIABLO (sparse generalized CCA, SGCCA; Tenenhaus *et al.*, 2014): WOVEN generalizes its supervised SUMCOR objective to the anchor-restricted block-missing setting and replaces its nonlinear iterative partial least squares (NIPALS) solver with a single eigendecomposition. Earlier complete-case integrators (iCluster+, Mo *et al.*, 2013; JIVE, Lock *et al.*, 2013) and recent block-missing predictors that impute missing blocks as preprocessing (Sui *et al.*, 2025) do not align a shared latent space; systematic review confirms existing methods remain inadequate for this regime (Hornung and Boulesteix, 2024).

The ADNI result is driven by the mechanism of the missingness. The modality most often absent is MRI, which more severely impaired patients disproportionately do not complete, so complete-case analysis excludes the most affected patients and biases downstream inference. Wherever missingness is driven by patient burden and functional capacity rather than random equipment failure, the same bias should be expected. By scoring these patients while preserving a more coherent latent space than complete-case analysis, WOVEN mitigates it directly.

Methodologically, the contribution is to show that the complete-case constraint, long treated as intrinsic to supervised multi-omics integration, is removable at no measured cost. DIABLO and its lineage force a choice between using a modality and keeping the patients who lack it; WOVEN removes that choice, scoring the entire cohort with equal or better discrimination and the same interpretable projection matrices. The advantage is largest precisely where it matters most, in clinical cohorts whose missing patients are the most severely affected, which positions WOVEN as a practical default for comparative effectiveness analyses that can no longer afford to discard them.

WOVEN has clear boundaries. Because all parameters are estimated from anchors, accuracy depends on the anchor fraction: below roughly 15% it degrades (Section 3.6), and single-modality subjects, which receive no cross-modal averaging, are the most sensitive. Prospective studies can guard against this by collecting one high-coverage base modality for every subject, or by staggering acquisition so a designated subset completes all modalities. WOVEN is also supervised: it requires labels on the anchor subjects, and without the supervision term it degenerates to unsupervised CCA (Section 2.5), so unlabeled exploratory analysis remains the domain of methods such as MOFA+ or IntegrAO. The latent dimension K is likewise user-specified; we fixed $K = 5$ to match the simulation ground truth, but in practice it can be chosen by the same anchor cross-validation used for the other hyperparameters.

Future work includes feature-level missingness within otherwise observed modalities, longitudinal cohorts where the anchor set shifts over time, sparse WOVEN variants for settings that demand interpretable loadings, and Bayesian uncertainty quantification for the projection; single-modality subjects, in particular, could be stabilized by smoothing their coordinates along the k-NN graph already built for regularization.

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Supplementary Information

Supplementary Methods S1: Simulator specifications

RNA-seq counts (ARMs A and B) were generated with SPsimSeq (Assefa *et al.*, 2020), a semi-parametric simulator that fits a Gaussian copula to a real RNA-seq reference and draws new samples preserving negative-binomial marginals and gene-gene correlations (4,047 genes after quality filtering). Methylation beta values (367 CpG sites), and for ARM B an additional 62 protein abundances, were generated jointly with group labels by InterSIM (Chalise *et al.*, 2016) with parameters estimated from the TCGA-OV (The Cancer Genome Atlas Ovarian Cancer) cohort. ARM C generates all three modalities jointly from the same reference, with strong between-group separation arising from the reference cohort structure. For ARM D, microbiome profiles were generated with MIDASim (He *et al.*, 2024), a two-step Gaussian copula simulator preserving taxon-taxon correlations, zero-inflation, and alpha diversity; metabolomics features were generated under the NorTA (Normal-to-Anything) framework (Mangnier *et al.*, 2025) with cross-modal associations spiked in using SparseDOSSA2 (Ma *et al.*, 2021). In the block-missingness step, each modality block is removed independently with the specified MCAR or MAR probability, subject to the constraint that every enrolled subject retains at least one observed modality.

Supplementary Methods S2: Solver details

Anchor latent scores are scaled so that $Z_v^T M_v Z_v = I_K$ (M-orthonormality). The recovery $W^{(v)} = X_a^{(v)T}(X_a^{(v)} X_a^{(v)T} + \epsilon I)^{-1} Z_v$ is the regularized minimum-norm least-squares solution; the ridge $\epsilon > 0$ ensures numerical stability when $p_v < n_a$ and the Gram matrix $X_a^{(v)} X_a^{(v)T}$ is rank-deficient, with a Moore-Penrose pseudoinverse used as a fallback. Only positive eigenvalues of P are retained; negative eigenvalues correspond to anti-correlated directions and are discarded. For $V = 2$, P has a single nonzero off-diagonal block, and the decomposition reduces to the singular value decomposition (SVD) of $M_1^{-1/2}(I + \gamma_Y K_Y)M_2^{-1/2}$, equivalent to dual supervised CCA. In the typical omics regime $p_v \gg n_a$, the Gram matrix $X_a^{(v)} X_a^{(v)T}$ is $n_a \times n_a$ and generically full rank, so the ridge ϵ governs numerical conditioning rather than invertibility; high feature dimension does not destabilize the recovery of $W^{(v)}$, and the binding resource is the anchor count n_a (addressed in Section 3.6) rather than p_v.

Additional supplementary materials: dual MCCA proof, Nystrom extension derivation, per-arm results tables (Supplementary Table S1), and sensitivity figures (Supplementary Figure 2).

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Conflict of Interest

None declared.

Data Availability

Simulation scripts, benchmark pipeline, and aggregate summary statistics are available at https://github.com/NathanBresette/woven_paper. The WOVEN R package, including all source code, documentation, and a built-in example dataset, is available at https://github.com/NathanBresette/woven (Bioconductor submission in progress). A stable archived version will be deposited at Zenodo upon acceptance. ADNI data are available to qualified researchers via application at adni.loni.usc.edu; individual-level data cannot be shared directly due to the ADNI data use agreement.